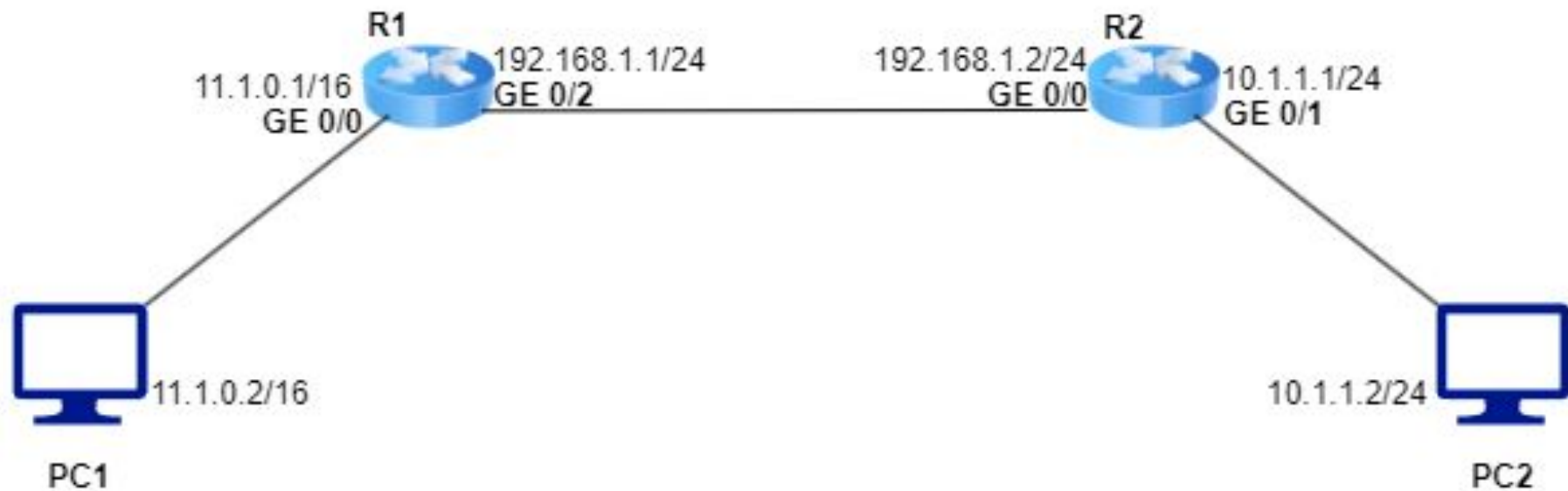


Demo - Network Setup and Route Configuration in Router

Network Setup



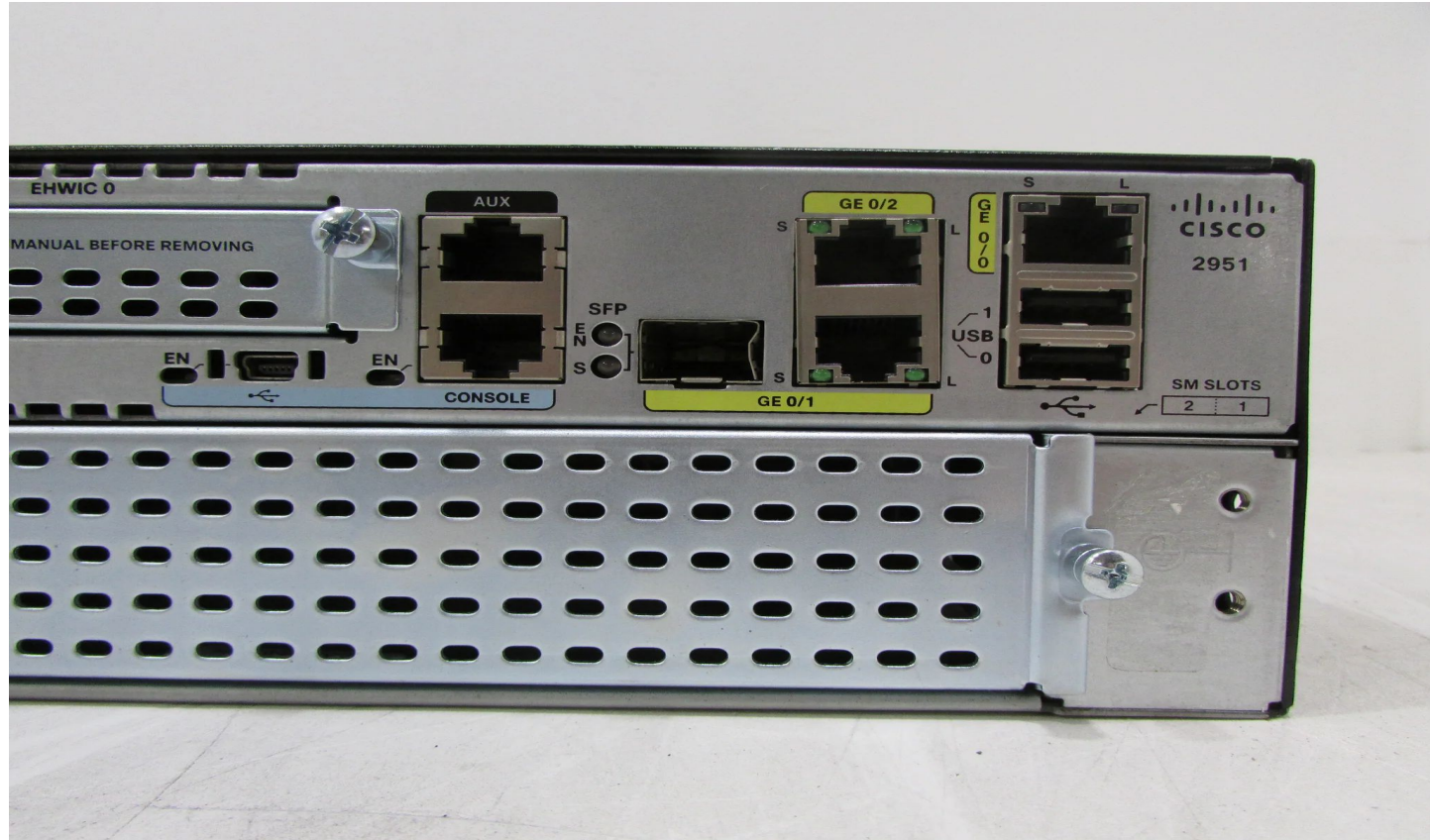
Network Setup - (Contd.)

- 2 Cisco Routers (Model 2900), 3 LAN cables, 1 Console cable(serial to USB) and 3 generic PCs (1 to connect to console port for router configuration).
- Connect one LAN cable from PC1 to GE 0/0 of R1.
- Similarly connect another LAN cable from PC2 to GE 0/1 of R2.
- Connect GE 0/2 of R1 with GE 0/0 of R2 using the third LAN.
- Laptop with Windows and Putty Installed for router configuration

Driver

1. Download **uc232a_windows_setup_v1.0.087.zip** from https://www.aten.com/global/en/supportcenter/info/downloads/?action=display_product&pid=575
2. Unzip and install the driver

Device Information - Ports



Cables



Name ?
Use Case ?



Name ?
Use Case ?



Name ?
Use Case ?

Network Setup -



Distribution of IP across respective interfaces -

Devices	IP	Subnet Mask	Gateway	Interfaces
PC1	11.1.0.2	255.255.0.0	11.1.0.1	Ethernet
PC2	10.1.1.2	255.255.255.0	10.1.1.1	Ethernet
Router (R1)	11.1.0.1	255.255.0.0		GE 0/0
Router (R1)	192.168.1.1	255.255.255.0		GE 0/2
Router (R2)	10.1.1.1	255.255.255.0		GE 0/1
Router (R2)	192.168.1.2	255.255.255.0		GE 0/0

Static IP allocation - (for PC1 & PC2)

PC1

Manual

IPv4

☒ On

IP address

11.1.0.2

Subnet mask

255.255.0.0

Gateway

11.1.0.1

Preferred DNS

0.0.0.0

Preferred DNS encryption

Unencrypted only

Alternate DNS

PC2

Editing wired connection 1

Connection name: Wired connection 2

General Ethernet 802.1x Security DCB IPv4 Settings IPv6 Settings

Method: Manual

Addresses

Address	Netmask	Gateway	
10.1.1.2	24	10.1.1.1	Add Delete

DNS servers:

Search domains:

DHCP client ID:

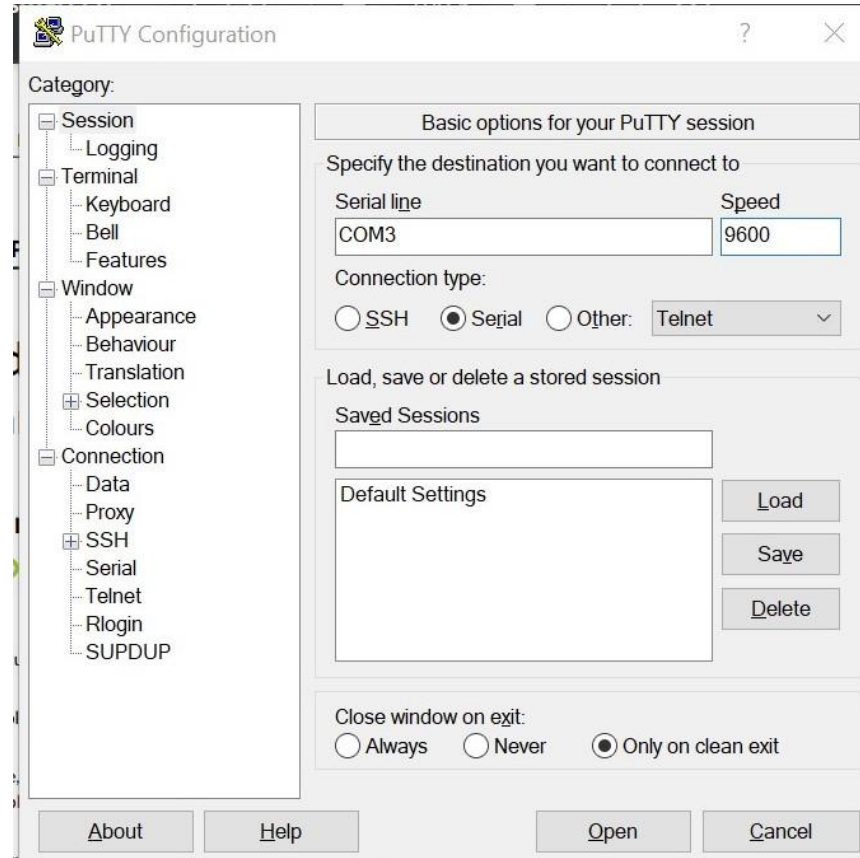
☒ Require IPv4 addressing for this connection to complete

Routes...

Cancel Save

Router Configuration (For R1) -

Configure PuTTY -



Router Configuration (For R1) -

Set Router Name-

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

Router Configuration (For R1) -

Assign IP address to interface GE 0/0

```
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface gigabitethernet 0/0
R1(config-if)#ip address 11.1.0.1 255.255.0.0
R1(config-if)#no shutdown
R1(config-if)#
```

Router Configuration (For R1) -

Assign IP address to interface GE 0/2

```
R1(config-if)#no shutdown  
R1(config-if)#interface gigabitethernet 0/2  
R1(config-if)#ip address 192.168.1.1 255.255.255.0  
R1(config-if)#no shutdown  
R1(config-if)#
```

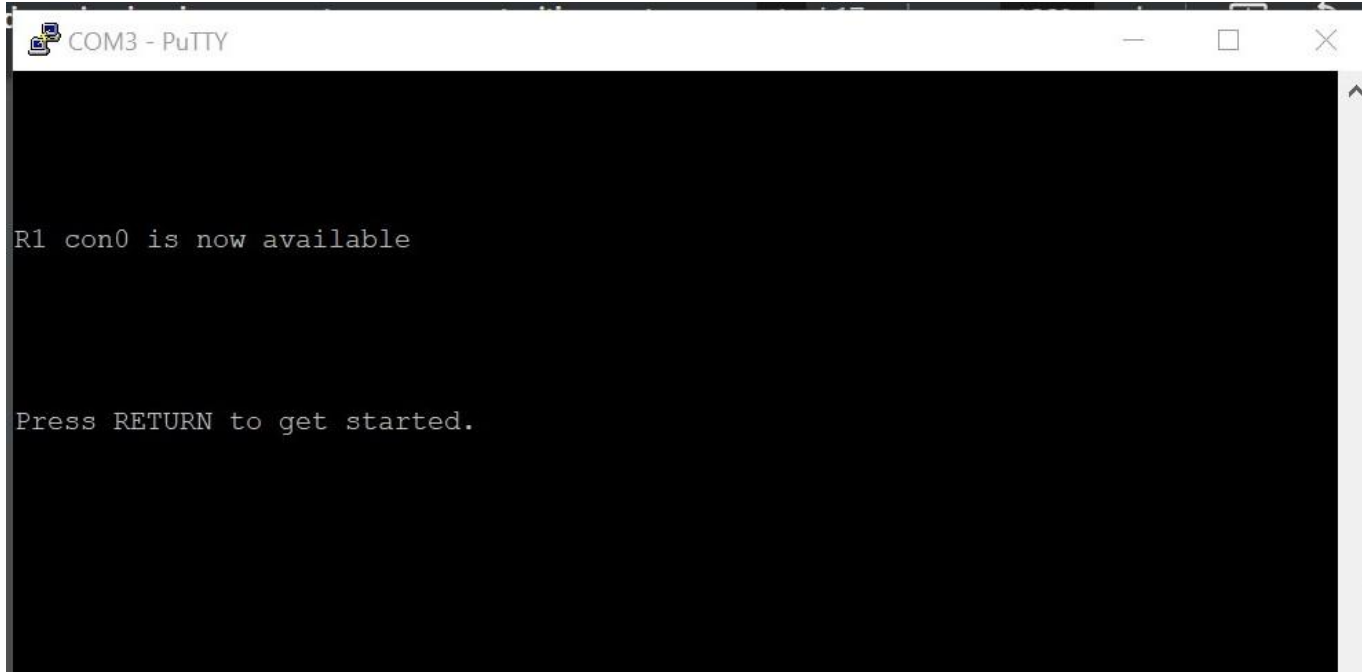
Router Configuration (For R1) -

Set static IP Route -

```
R1(config-if)#exit  
R1(config)#ip route 10.1.1.0 255.255.255.0 192.168.1.2  
R1(config)#
```

Router Configuration (For R1) -

Switch from the Configuration mode and move to user mode using exit command



```
COM3 - PuTTY

R1 con0 is now available

Press RETURN to get started.
```

Router Configuration (For R2) -

Plug in the console cable to Router R2 and assign IP address to interface GE 0/1

```
R2(config-if)#interface gigabitethernet 0/1
R2(config-if)#ip address 10.1.1.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#
*Oct 14 10:13:41.103: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
*Oct 14 10:13:50.271: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
*Oct 14 10:13:51.271: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```


Router Configuration (For R2) -

Assign IP address to interface GE 0/0

```
R2(config)#interface gigabitethernet 0/0
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#
*Oct 14 10:11:03.827: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to down
*Oct 14 10:11:07.271: %LINK-3-UPDOWN: Interface GigabitEthernet0/0, changed state to up
*Oct 14 10:11:08.271: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

Router Configuration (For R2) -

Set static IP route -

```

R2(config-if)#exit
R2(config)#ip route 11.1.0.0 255.255.255.0 192.168.1.1
R2(config)#
```

Testing-

```
C:\WINDOWS\system32>ping 10.1.1.2
```

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time=1ms TTL=62

Reply from 10.1.1.2: bytes=32 time=1ms TTL=62

Reply from 10.1.1.2: bytes=32 time=1ms TTL=62

Reply from 10.1.1.2: bytes=32 time=1ms TTL=62

Ping statistics for 10.1.1.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 1ms, Average = 1ms

```
C:\WINDOWS\system32>
```

```
oem@dell-desktop:~$ ping 11.1.0.2
PING 11.1.0.2 (11.1.0.2) 56(84) bytes of data:
64 bytes from 11.1.0.2: icmp_seq=1 ttl=126 time=1.37 ms
64 bytes from 11.1.0.2: icmp_seq=2 ttl=126 time=1.79 ms
64 bytes from 11.1.0.2: icmp_seq=3 ttl=126 time=1.63 ms
64 bytes from 11.1.0.2: icmp_seq=4 ttl=126 time=1.46 ms
64 bytes from 11.1.0.2: icmp_seq=5 ttl=126 time=1.22 ms
64 bytes from 11.1.0.2: icmp_seq=6 ttl=126 time=1.32 ms
64 bytes from 11.1.0.2: icmp_seq=7 ttl=126 time=1.63 ms
64 bytes from 11.1.0.2: icmp_seq=8 ttl=126 time=1.10 ms
64 bytes from 11.1.0.2: icmp_seq=9 ttl=126 time=2.51 ms
64 bytes from 11.1.0.2: icmp_seq=10 ttl=126 time=2.41 ms
64 bytes from 11.1.0.2: icmp_seq=11 ttl=126 time=2.67 ms
64 bytes from 11.1.0.2: icmp_seq=12 ttl=126 time=2.15 ms
64 bytes from 11.1.0.2: icmp_seq=13 ttl=126 time=2.15 ms
^C
--- 11.1.0.2 ping statistics ---
13 packets transmitted, 13 received, 0% packet loss, time 12022ms
rtt min/avg/max/mdev = 1.101/1.804/2.677/0.504 ms
oem@dell-desktop:~$
```

Testing(Trace Route)

```
C:\WINDOWS\system32>tracert 10.1.1.2
```

```
Tracing route to 10.1.1.2 over a maximum of 30 hops
```

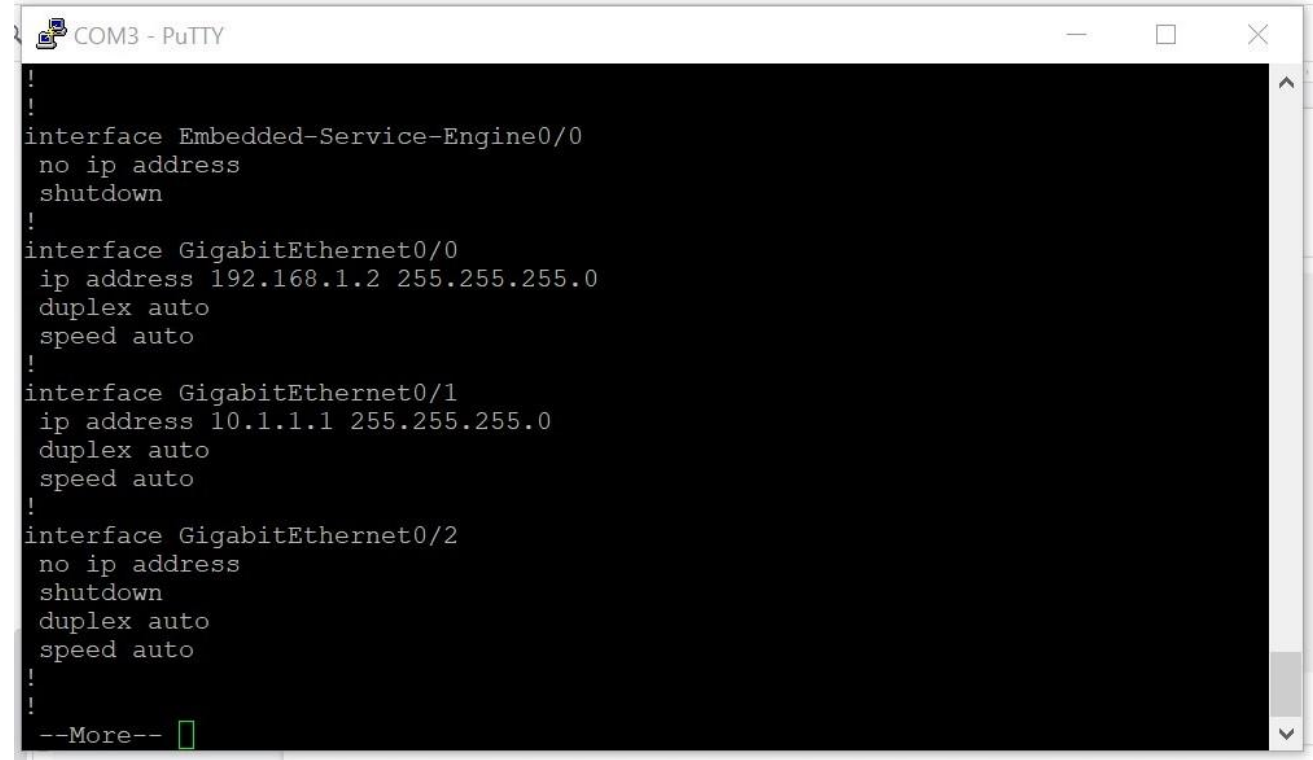
1	1 ms	<1 ms	<1 ms	11.1.0.1
2	1 ms	<1 ms	<1 ms	192.168.1.2
3	2 ms	2 ms	1 ms	10.1.1.2

```
Trace complete.
```

```
C:\WINDOWS\system32>
```

Testing (Check router configuration using 'sh run' command)

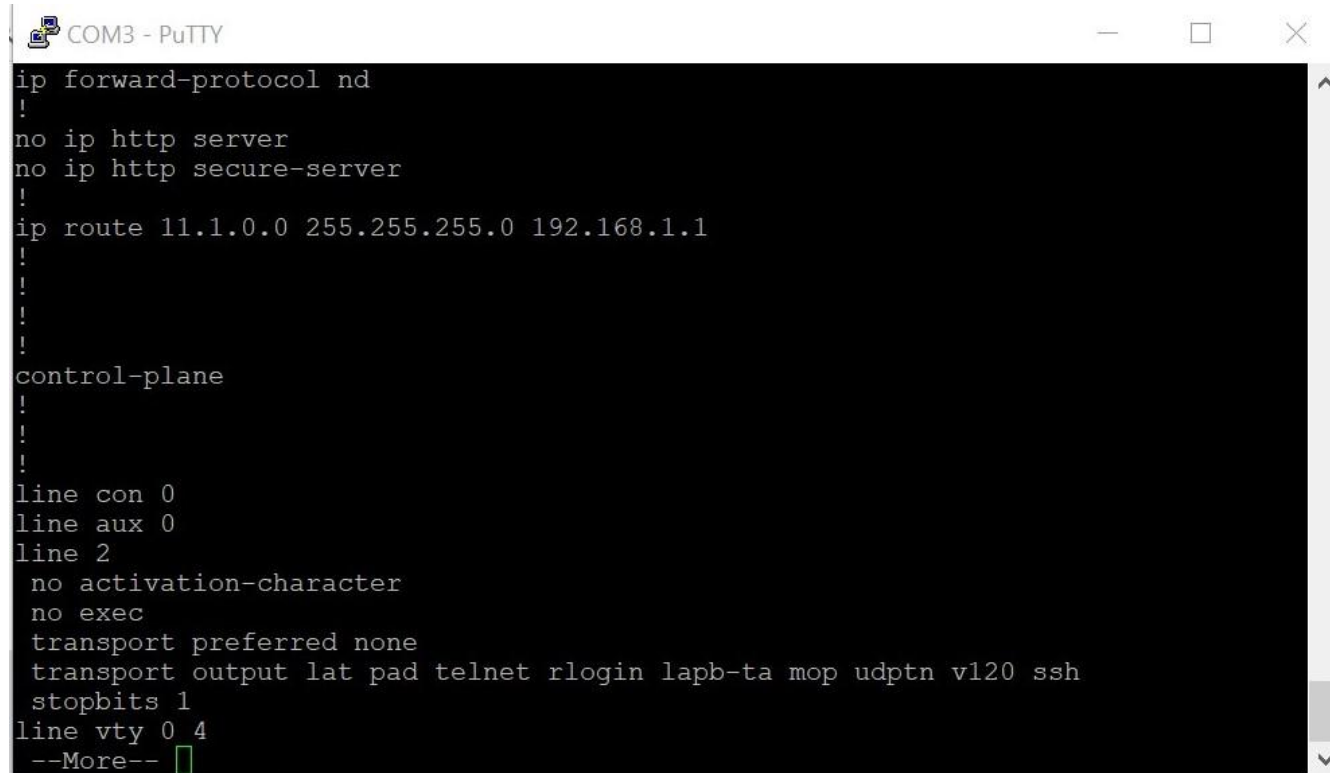
For IP -

A screenshot of a PuTTY terminal window titled 'COM3 - PuTTY'. The window has a black background with white text. The text shows a series of configuration commands for three interfaces: Embedded-Service-Engine0/0, GigabitEthernet0/0, GigabitEthernet0/1, and GigabitEthernet0/2. The commands include 'interface', 'ip address', 'duplex auto', 'speed auto', and 'shutdown'. The configuration is enclosed in '!' markers. At the bottom, there is a '--More--' prompt followed by a green cursor.

```
!
!  
interface Embedded-Service-Engine0/0  
  no ip address  
  shutdown  
!  
interface GigabitEthernet0/0  
  ip address 192.168.1.2 255.255.255.0  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/1  
  ip address 10.1.1.1 255.255.255.0  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/2  
  no ip address  
  shutdown  
  duplex auto  
  speed auto  
!  
!  
--More--
```

Testing (Check router configuration)

For static Route -

A screenshot of a PuTTY terminal window titled "COM3 - PuTTY". The window has standard window controls (minimize, maximize, close) in the top right corner. The terminal displays a series of configuration commands for a router. The commands are: "ip forward-protocol nd", "no ip http server", "no ip http secure-server", "ip route 11.1.0.0 255.255.255.0 192.168.1.1", "control-plane", "line con 0", "line aux 0", "line 2", "no activation-character", "no exec", "transport preferred none", "transport output lat pad telnet rlogin lapb-ta mop udptn v120 ssh", "stopbits 1", "line vty 0 4", and "--More--". A green cursor is visible at the end of the "--More--" line.

```
COM3 - PuTTY
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
ip route 11.1.0.0 255.255.255.0 192.168.1.1
!
!
!
!
control-plane
!
!
!
!
line con 0
line aux 0
line 2
  no activation-character
  no exec
  transport preferred none
  transport output lat pad telnet rlogin lapb-ta mop udptn v120 ssh
  stopbits 1
line vty 0 4
--More--
```

QnA

1. How Many Networks are setup in this scenario?
2. Is the routing table populated dynamically?
3. What should traceroute must provide?
4. Which type of IP address is used in configuration?
5. IP address of which class is used during configuration?
6. Used IP address is classless or classful?
7. What is the network ID of PC1 and PC2?

Thank you